International
Max Planck Research School
for language sciences

Programming for Research 2025

Esam Ghaleb & Loïs Dona
2.5 ECs

COURSE OVERVIEW

COURSE DESCRIPTION & INSTRUCTOR PROFILES

GOALS

Whether you are a computational researcher or not, chances are high that you will have to do programming one way or another. This course covers the basics of programming at various levels of implementation, tailored to language scientists. Although focused on Python, the course provides students with domain-general programming skills that can be used in their research. The lectures alternate between more theoretical lectures and more applied "practical" lectures.

We plan to follow this course with an introductory course to machine learning basics. Hence, this course (or basic programming skills) will be prerequisite.

LEARNING OUTCOME

The course will teach students to:

- Install and use programming tools for Python
- Understand the basics of programming concepts (e.g., variables, arrays, objects, functions, etc.) and programming paradigms (e.g., object-oriented, functional, etc.)
- Understand how to process data, from loading to visualization
- Get an overview of various packages, functions, and tools that are useful for language science
- Write clear, well-structured, and well-commented code that is suitable for sharing with other academics as well as for publication alongside journal articles
- Implement basic experiment designs in PsychoPy
- Use documentation, online communities (e.g., StackOverflow), debuggers, and AI assistants to troubleshoot problems

PRE-REQUISITES

Familiarity with programming is **not required** but is invariably helpful. The instructor will provide documentation and resources for each lecture, many of which will be optional reading material for further exploration of a particular topic. Beginners are strongly encouraged to explore these materials.

ABOUT THE INSTRUCTOR

Esam Ghaleb is a research scientist at the MLD. His education and research background are in computer science and AI. Programming is a tool for his research and development of machine learning and computer vision models to study affective and communicative behaviors. His favorite programming practice is debugging. According to Esam, writing efficient and readable code is the best practice and rule in daily programming tasks.

Loïs Dona is a PhD student in the MLD and LEADS group. Her background is in AI, with a strong focus

on language and cognitive computational modeling. Fun fact: She started her AI studies without ever having programmed before! Learning programming might seem daunting to some of you, but she hopes to give you the tips and tricks to help you become more confident in your programming skills.

CONTACT INFORMATION
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FORMAT, SCHEDULE & CONTENT OUTLINE

Format & preparations

The course will consist of **12 lectures** of **120 minutes** each. You must attend **at least 10 sessions** (including doing the assignments for those sessions) to obtain ECs. The lecture format follows a workshop-like template, sometimes with cooperative programming in class. During lectures, there will be plenty of space for asking questions. There will be assignments after half of the lectures.

You will need a laptop for this course. You must have sufficient file system privileges to install the software. You should use a machine with an admin account. Several software packages and/or programs will be necessary. Almost all of these will be installed in Session 0. **All participants are expected to attend Session 0**.

Content outline & schedule

Session 0 Course Introduction and Preparation Introduction to Python, Visual Studio Code and Jupyter Notebook. As well as an introduction to Python Syntax and Concepts.	**Monday Feb 3** **10:00 - 12:00** **MPI 236**
Session 1 Exploring Fundamentals of Programming Explore key programming concepts like data types and data structures. These concepts are the foundations of algorithmic thinking and are helpful for every programming language. We will also introduce Python tools such as Numpy, commonly used for scientific research.	**Monday Feb 10** **10:00 - 12:00** **MPI 236**
Session 2 Mastering Functions, Logic, and Loops: The Core Building Blocks of Programming. Learn core concepts that help you make your code more efficient and reusable: functions, logical operators, and loops.	**Monday Feb 17** **10:00 - 12:00** **MPI 236**
Session 3 Advanced Data Types: Objects and Classes Explore more advanced data types in object-oriented programming with objects and classes.	**Monday Feb 24** **10:00 - 12:00** **MPI 236**
Session 4 Writing Readable and Comprehensible Code: Best Practices for Academics This session introduces best practices for writing clean, structured, and well-documented code tailored to research workflows. Learn techniques to improve your projects' consistency, collaboration, and long-term usability.	**Monday Mar 3** **10:00 - 12:00** **MPI 236**
Session 5 Practical Session I: Hands-On Programming Fundamentals This session is meant to solidify the skills that you've learnt so far, by putting your programming skills to the test in this hands-on session. This session focuses on applying various data types, functions, logic, and loops through guided exercises.	**Monday Mar 10** **10:00 - 12:00** **MPI 236**
Session 6 Data Processing, Visualization, and File Handling: From Loading to Insight—> already use Pandas for this. Master file handling, data processing, and visualization to gain insight into your data. Learn to load, clean, and analyze data from various formats and create impactful visualizations.	**Monday Mar 17** **10:00 - 12:00** **MPI 236**
Session 7 Essential Packages and Tools for Language Science: Pandas, NLTK, and SpaCy Discover essential tools for language science. This session covers Pandas for data analysis, NLTK, and SpaCy for text processing. Learn to use packages, preprocess text and analyze linguistic data.	**Monday, Mar 24** **10:00 - 12:00** **MPI 236**
Session 8 Introduction to Experiment Design with Python: PsychoPy	**Monday, Mar 31**

Learn the essentials of experiment design in PsychoPy, a Python-powered framework for programming experiments. This session covers creating stimuli, controlling experimental flow, and collecting data.	**10:00 - 12:00** **MPI 236**
Session 9 Advanced Debugging: Using ChatGPT, Stack Overflow, and GitHub Copilot Master debugging with tools like ChatGPT, Stack Overflow, and GitHub Copilot. Learn to ask better questions, decode errors, and implement fixes to enhance your coding workflow and productivity. We will also highlight the perks and pitfalls of these tools.	**Monday, Apr 7** **10:00 - 12:00** **MPI 236**
Session 10 Practical Session II: Hands-On Data Processing and Visualization This session is meant to solidify the skills that you've learnt so far, by putting your programming skills to the test in this hands-on session. This session focuses on file handling, data processing and visualization using various Python packages.	**Monday, Apr 14** **10:00 - 12:00** **MPI 236**
Session 11 Mystery Guest Lecture! To be announced	**Thursday, Apr 24** **10:00 - 12:00** **MPI 236**

ATTENDANCE POLICY, REGISTRATION & FEEDBACK

This course will consist of twelve 120-minute lectures for a total of 24 lecture hours. In addition, there will be approximately three hours of assignment work after 5 lectures, for a total of 15 assignment hours.

The total time investment is approximately 39 hours or 2.5 EC.
Register **here** **(Max 15 participants).**
Priority will be given to those earning ECs (i.e., full attendance and completion of assignments).

We expect participants to ask many questions and make a serious effort with the homework assignments. We will be available weekly to answer questions and help solve problems.

IMPRS policy: To earn full credits, participants must fully attend the minimum required number of sessions (10) and complete the corresponding assignments.

Course-specific policy: Attending more than the required lectures will not change the number of ECs earned.